

Jonathan Cruz

PRINCIPAL SOFTWARE ENGINEER · AI-AUGMENTED ENGINEERING

📍 SÃO PAULO, BRAZIL

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PROFILE

"Senior engineer with over 10 years building consumer and B2B platforms across web and mobile, with a strong focus on portfolio-level engineering excellence. Daily practitioner of AI-augmented engineering — sub-agent orchestration, spec-driven workflows, and agentic code generation/review pipelines. Deep TypeScript expertise across the stack with React/React Native, NestJS, Prisma, and Nx monorepos. Track record of mentoring engineers from junior to senior, modernizing legacy systems, and shipping reliable solutions in regulated and high-availability environments."

PROFESSIONAL EXPERIENCE

SELECTED ROLES & CONTRIBUTIONS

FINTECH FOR VC FUND
ADMINISTRATION

Sydecar

FULL-STACK SOFTWARE
ENGINEER

AI-
Augmented
Engineering
AGENTIC WORKFLOWS

Daily user of agentic coding tools (Claude Code) with sub-agent orchestration, custom skills, project-level CLAUDE.md specifications, and pre/post-tool hooks. Apply spec-driven workflows for code generation, test scaffolding, code review, and architectural exploration — pairing AI execution layers with knowledge-centric project conventions to maintain quality at high iteration speed.

AI ENGINEERING AGENTIC WORKFLOWS SPEC-DRIVEN SUB-AGENTS

Nx
Monorepo
CROSS-TEAM
ARCHITECTURE

Operate inside an Nx monorepo spanning dozens of NestJS libraries and React/Next.js apps. Drive cross-library architectural decisions (ADRs), enforce module boundaries, and coordinate changes across product teams to keep delivery flowing without blocking parallel workstreams.

NX MONOREPO ADR NESTJS

Complex
Backend
Systems
POSTGRES

Designed and implemented Enhanced Due Diligence with LP funding blocks, UBO verification, FinCEN integration, and CTA compliance. Built race-resistant flows with re-check inside to handle concurrent edits (409 EMAIL_NOW_EXISTS). Contributed to a proprietary double-entry ledger with append-only entries, deterministic idempotency keys, and event-driven posting.

POSTGRES PRISMA CONCURRENCY COMPLIANCE

Fault
Tolerance
EVENT-DRIVEN

Standardized event delivery using BullMQ Pro with exponential backoff (5 retries / 3s base; 20 retries / 10s base in pubsub). Outbox pattern: persist resolution first, enqueue after — jobs idempotent and survive enqueue failure. Global per-deal ordering via Redis INCR counters with deterministic predecessor lookup ().

BULLMQ REDIS OUTBOX EVENT-DRIVEN

External
Integrations
CROSS-ORG
COLLABORATION

Built bidirectional QuickBooks invoice sync engine with state validation, auto-sync on approval, full E2E suite with Playwright and Cucumber BDD. Worked on Plaid Link integration with processor tokens powering ACH via Treasury Prime, including mock banking engine for local sandbox.

PLAID TREASURY PRIME QUICKBOOKS PLAYWRIGHT

Type-safe
Contracts
DX EXCELLENCE

Designed (ADR) and implemented Billing V2 with quarterly fee scheduler and billing model versioning. Migrated legacy controllers (auth, onboarding, admin) to type-safe ts-rest contracts with Zod schemas — improving developer experience and contract reliability end-to-end.

TS-REST ZOD TYPESCRIPT

MAY 2023 — PRESENT
SAN FRANCISCO, CA (REMOTE)

B2B OMNICHANNEL WHOLESALE

Zax

SR SOFTWARE ENGINEER

Omnichannel
Platform
MOBILE + WEB

Built from scratch a wholesale ordering app integrating mobile and web channels for business partners, with credit-limit validation and order-to-cash flows, using React Native, TypeScript, and GraphQL.

REACT NATIVE TYPESCRIPT GRAPHQL

Scalable
Cloud
AWS

Deployed full cloud infrastructure on AWS (EC2, ECS, S3, Lambda, VPCs) with serverless web apps using React and Next.js. Containerized services with Docker for consistent dev-to-prod parity.

AWS NEXT.JS SERVERLESS DOCKER

Engineering
Excellence
CI/CD

Configured end-to-end delivery pipeline with GitHub Actions, Fastlane, and Code Climate for automated quality gates. Reduced release cycle from weeks to days.

GITHUB ACTIONS FASTLANE DOCKER

JAN 2019 — MAY 2023
SÃO PAULO

CONSUMER MOBILE + WEB

Netlolo

SR SOFTWARE ENGINEER

SEP 2018 — JAN 2019

SÃO PAULO

Consumer Product

REACT NATIVE

Built a mobile and web product bringing cryptocurrency to mainstream users with React Native and TypeScript, integrating Web3 smart contracts on Ethereum. Demonstrated interest in researching and shipping alternative/emerging web technologies.

REACT NATIVE

TYPESCRIPT

WEB3

Real-time Systems

SDK

Developed a chat SDK for businesses using React Native and React/Redux, powered by XMPP and a Java backend for low-latency real-time messaging across multiple client integrations.

REACT

REDUX

XMPP

JAVA

Zero Downtime

Built CI/CD automations ensuring continuous deployment with no service interruption. PostgreSQL with Hasura as the GraphQL layer.

POSTGRESQL

HASURA

CI/CD

CONSUMER HEALTHCARE

Escale

SR SOFTWARE ENGINEER

JAN 2018 — SEP 2018

SÃO PAULO

Mentorship

ENGINEERING CULTURE

Led internal Tech Talks on RESTful design, Functional Programming, and Microservices architecture, coaching engineers across product teams. Established patterns for service boundaries and API contracts adopted across the engineering org.

MENTORSHIP

TECH TALKS

MICROSERVICES

Microservices Architecture

MULTI-DB

Built the product foundation with React, Redux, and a Node.js/TypeScript backend. Selected appropriate persistence per service responsibility — MySQL, PostgreSQL, and MongoDB — driving DB design and tuning decisions.

REACT

NODE.JS

MONGODB

CIRCLECI

FINTECH

PraValer

PL SOFTWARE ENGINEER

JUN 2017 — JAN 2018

SÃO PAULO

Legacy Modernization

NODE

Migrated legacy PHP origination platform to a Node.js/Express backend for a student loans product with governmental funding, modernizing API surface while preserving downstream consumer compatibility.

NODE.JS

EXPRESS

LEGACY MIGRATION

Database

PERFORMANCE TUNING

Restructured PostgreSQL transactional tables with improved schemas and indexes, eliminating N+1 queries and reducing query times by an order of magnitude. MongoDB for request logs and unstructured data.

POSTGRESQL

MONGODB

QUERY OPTIMIZATION

CORE SKILLS

AI ENGINEERING

Agentic workflows, Sub-agent orchestration, Spec-driven development, Prompt engineering, AI-augmented code review and testing

BACKEND

TypeScript, NestJS, Node.js, Prisma, PostgreSQL, ts-rest, Zod, GraphQL, Event-driven architecture

FRONTEND

React, Next.js, React Native, RxJS, TanStack Query, HTML, CSS preprocessors

RELIABILITY

BullMQ, Outbox pattern, Idempotency, Redis, Datadog

INFRASTRUCTURE

AWS, GCP, Docker, CI/CD, Nx Monorepo, Git

LEADERSHIP

Mentoring engineers (junior → senior), ADRs, Cross-team coordination, Tech Talks

LANGUAGES

PORTUGUESE

Native

ENGLISH

C1/C2

EDUCATION

Anhanguera

BACHELOR OF TECHNOLOGY (B.TECH.) IN INFORMATION TECHNOLOGY · JAN 2016 — JUL 2018